SIMATS ENGINEERING

ASSESSMENT TOOL 3 – Assignment 2

COURSE NAME: STATISTICS WITH R PROGRAMMING **COURSE CODE: ITA04**

COURSE OUTCOME COVERED

CO3: *Construct and manipulate data frames using reshaping, merging, and file I/O operations in R to prepare structured datasets for analysis.. (BL3, BL5)*

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks: 100**

Annexure

Assignment 2

Code of Conduct:

I RAJA RAJESHWARI M (192421253) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
RAJA RAJESHWARI M
Signature of the student:

Assignment 2:

- **Generate 5 Questions for your Capstone Project module**
- **Apply the formulae and use the relevant graph to visualize it**
- **Upload the file in Github link**
- **Save the 5 Questions and Gihub link in the Excell sheet**

Food Waste Prediction

1. **How can Machine Learning algorithms predict food waste using historical consumption, customer demand, weather, and inventory data?**
   To identify relationships between factors such as customer demand, weather, and the
   amount of food wasted**.**
   Statistical Test : Chi-Square Test of Independence
   Formula: Pearson Correlation Coefficient

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

   Purpose: To identify whether higher customer demand is associated with higher food waste.
   CODE: # Q1: Association Between Food Category and Food Waste

```
# Create data with multiple observations
food <- matrix(c(
  40, 10,
  35, 15,
  28, 22,
  32, 18,
  20, 30,
  25, 25,
  45, 15,
  18, 32
), nrow = 8, byrow = TRUE)

rownames(food) <- c("Vegetables", "Fruits", "Dairy",
                    "Bakery", "Meat", "Grains",
                    "Seafood", "Leftovers")

colnames(food) <- c("Wasted", "Not_Wasted")

# Display data
print(food)

# Chi-square test
test <- chisq.test(food)
print(test)

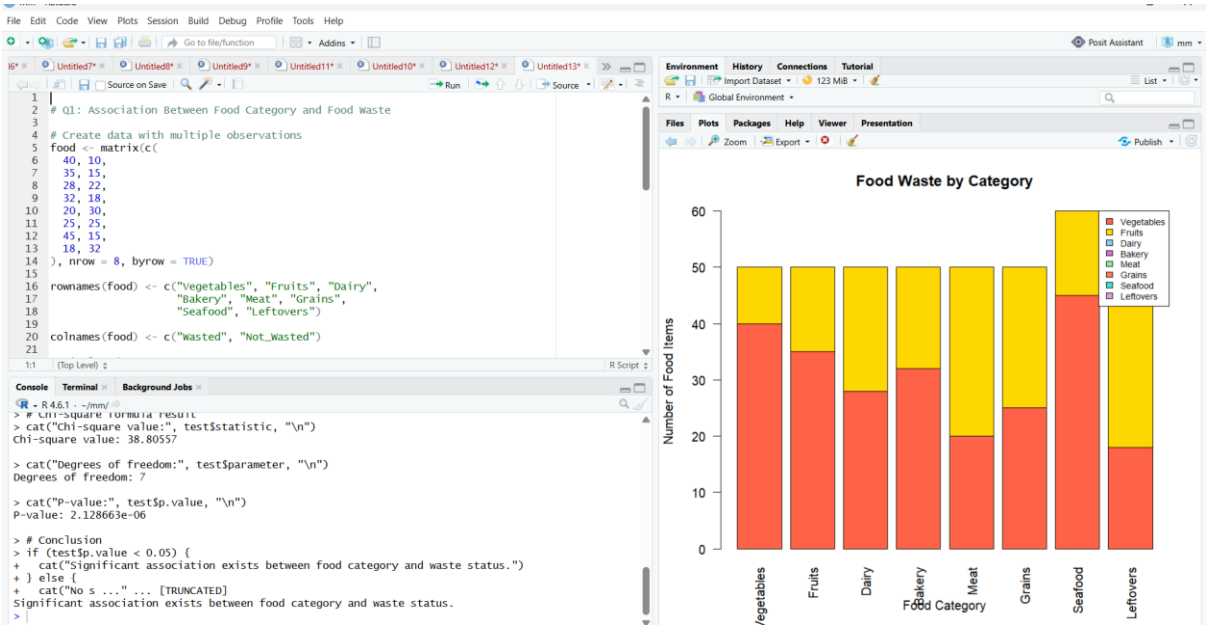
# Stacked bar chart with unique colours
barplot(t(food),
        beside = FALSE,
        col = c("tomato", "gold", "skyblue",
                "orchid", "lightgreen", "coral",
                "turquoise", "plum"),
```

```
        main = "Food Waste by Category",
        xlab = "Food Category",
        ylab = "Number of Food Items",
        border = "black",
        las = 2)

legend("topright",
       legend = rownames(food),
       fill = c("tomato", "gold", "skyblue",
                "orchid", "lightgreen", "coral",
                "turquoise", "plum"),
       cex = 0.7)

# Chi-square formula result
cat("Chi-square value:", test$statistic, "\n")
cat("Degrees of freedom:", test$parameter, "\n")
cat("P-value:", test$p.value, "\n")

# Conclusion
if (test$p.value < 0.05) {
  cat("Significant association exists between food category and waste status.")
} else {
  cat("No significant association exists between food category and waste status.")
}
```



2. **How can a Food Waste Prediction system help restaurants, supermarkets, and households reduce food wastage and costs?**
   Statistical Test :t-test
   Formula:t=sd/nd‾
   Visualization: Bar Chart

1. X-axis: Before System and After System
2. Y-axis: Average Food Waste (kg)
3. Two bars compare the average waste before and after implementing the system.

Purpose: To determine whether average food waste changes significantly after implementing the system.

CODE: # Q2: Impact of Food Waste Prediction System

```
days <- 1:7

before <- c(45, 50, 42, 48, 55, 46, 52)
after <- c(30, 32, 28, 35, 38, 31, 34)

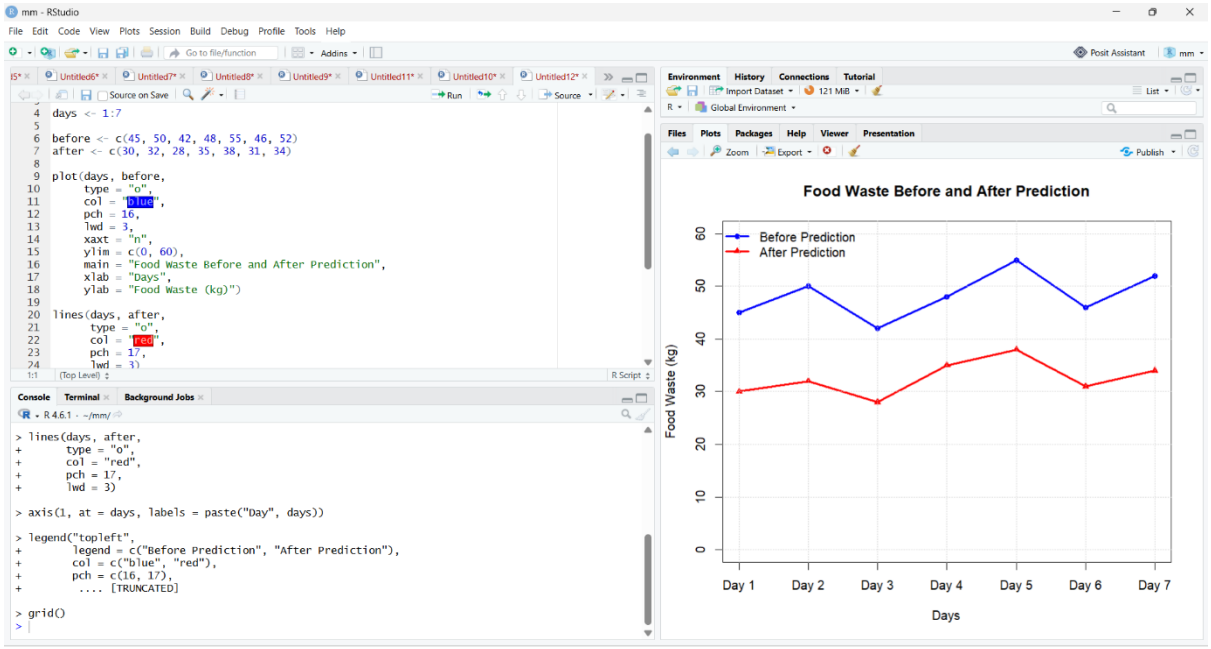
plot(days, before,
    type = "o",
    col = "blue",
    pch = 16,
    lwd = 3,
    xaxt = "n",
    ylim = c(0, 60),
    main = "Food Waste Before and After Prediction",
    xlab = "Days",
    ylab = "Food Waste (kg)")

lines(days, after,
     type = "o",
     col = "red",
     pch = 17,
     lwd = 3)

axis(1, at = days, labels = paste("Day", days))

legend("topleft",
      legend = c("Before Prediction", "After Prediction"),
      col = c("blue", "red"),
      pch = c(16, 17),
      lwd = 3,
      bty = "n")

grid()
```



3.Is the proportion of food orders wasted significantly different from the expected wastage rate of 10%?

Statistical Test: One-Way ANOVA

Formula:

$$\hat{p} = \frac{x}{n}$$

CODE:

```
# Q3: Food Wastage Rate vs Target

food <- c("Vegetables", "Fruits", "Dairy",
          "Bakery", "Meat", "Grains")
```

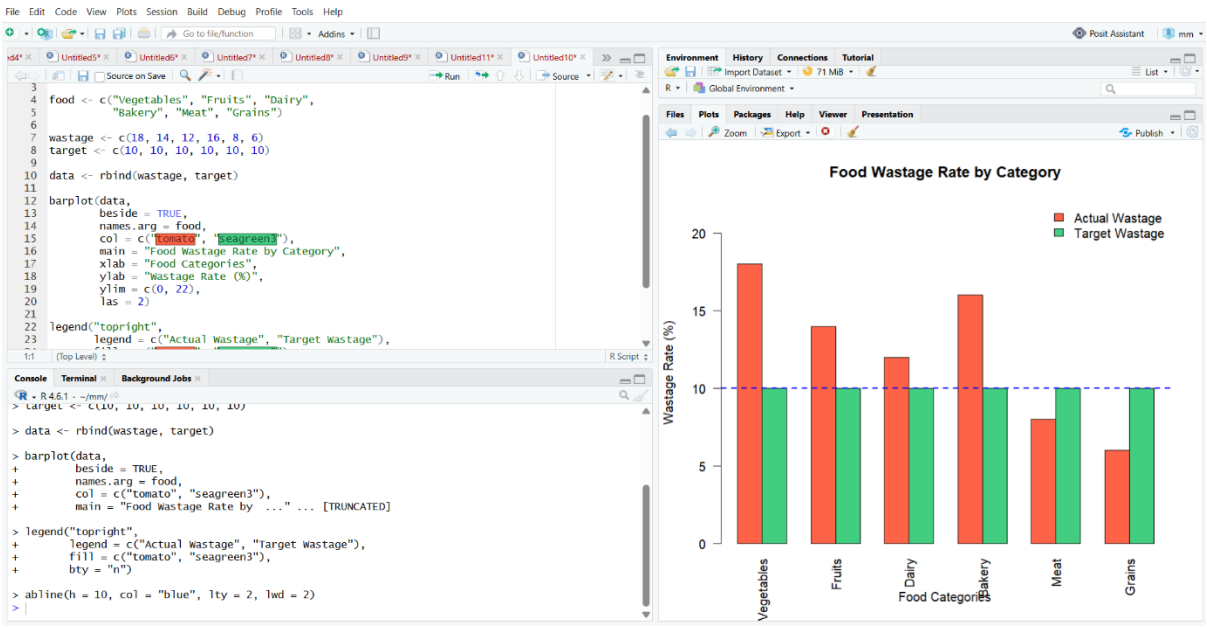
```
wastage <- c(18, 14, 12, 16, 8, 6)
target <- c(10, 10, 10, 10, 10, 10)

data <- rbind(wastage, target)

barplot(data,
        beside = TRUE,
        names.arg = food,
        col = c("tomato", "seagreen3"),
        main = "Food Wastage Rate by Category",
        xlab = "Food Categories",
        ylab = "Wastage Rate (%)",
        ylim = c(0, 22),
        las = 2)

legend("topright",
       legend = c("Actual Wastage", "Target Wastage"),
       fill = c("tomato", "seagreen3"),
       bty = "n")

abline(h = 10, col = "blue", lty = 2, lwd = 2)
```



4.Is there a significant association between food category and whether food is wasted?

Statistical Test: Pearson Correlation Test
Formula:
$\chi 2=\sum E(O-E)2$
Purpose: To determine whether food category and waste status are statistically associated.
CODE:

```
# Food Waste Analysis using One-Way ANOVA

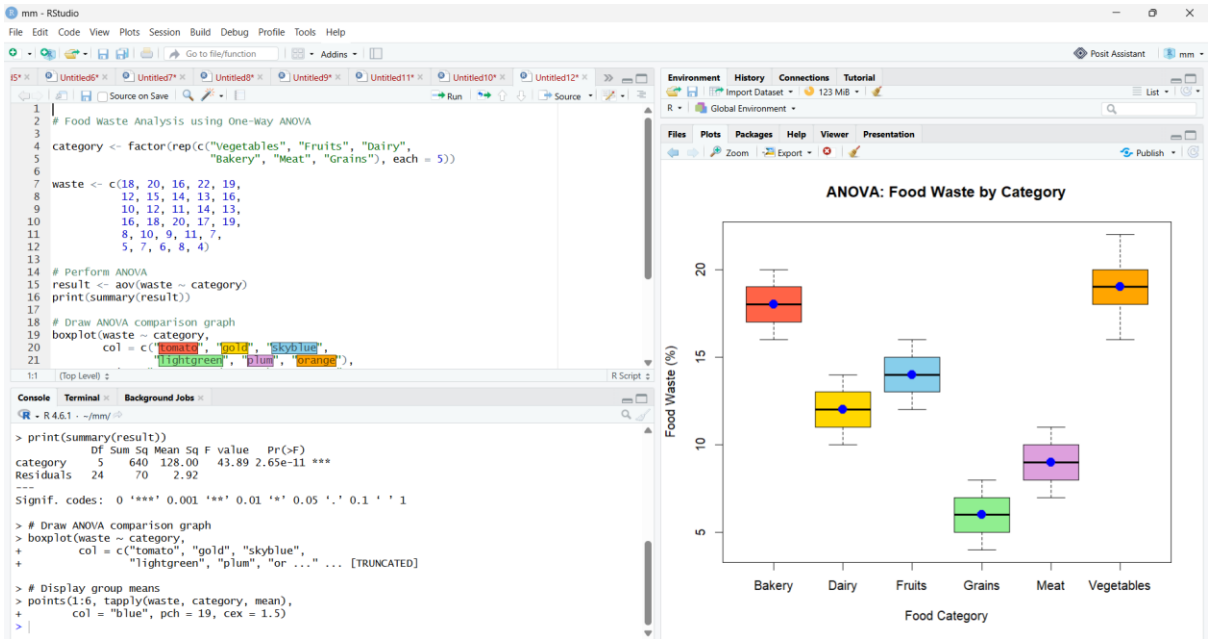
category <- factor(rep(c("Vegetables", "Fruits", "Dairy",
                         "Bakery", "Meat", "Grains"), each = 5))

waste <- c(18, 20, 16, 22, 19,
           12, 15, 14, 13, 16,
           10, 12, 11, 14, 13,
           16, 18, 20, 17, 19,
           8, 10, 9, 11, 7,
           5, 7, 6, 8, 4)

# Perform ANOVA
result <- aov(waste ~ category)
print(summary(result))

# Draw ANOVA comparison graph
boxplot(waste ~ category,
        col = c("tomato", "gold", "skyblue",
                "lightgreen", "plum", "orange"),
        main = "ANOVA: Food Waste by Category",
        xlab = "Food Category",
        ylab = "Food Waste (%)",
        border = "black")

# Display group means
points(1:6, tapply(waste, category, mean),
       col = "blue", pch = 19, cex = 1.5)
```



5. Does average food waste differ among restaurants, supermarkets, and households?
Statistical Test: One-way ANOVA
Formula: F=MSW/MSB
CODE:
Q5: Compare Food Waste Across Different Settings

days <- 1:7

restaurant <- c(25, 30, 28, 35, 32, 38, 34)
supermarket <- c(18, 22, 20, 24, 26, 23, 28)
household <- c(12, 15, 14, 18, 16, 20, 19)

waste_data <- rbind(restaurant, supermarket, household)

Create stacked area chart

```
area_colors <- c("tomato", "deepskyblue", "gold")

matplot(days, t(waste_data),
        type = "n",
        ylim = c(0, 100),
        xaxt = "n",
        main = "Food Waste Across Different Settings",
        xlab = "Day",
        ylab = "Food Waste (kg)")

axis(1, at = days, labels = paste("Day", days))

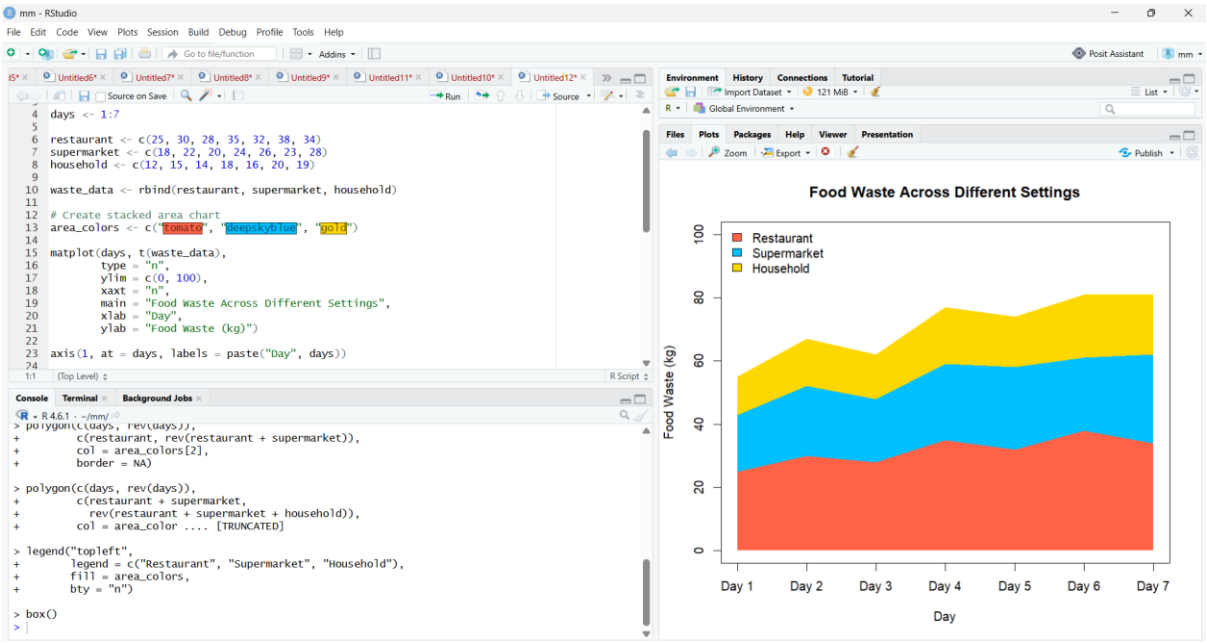
polygon(c(days, rev(days)),
        c(rep(0, length(days)),
          rev(restaurant)),
        col = area_colors[1],
        border = NA)

polygon(c(days, rev(days)),
        c(restaurant, rev(restaurant + supermarket)),
        col = area_colors[2],
        border = NA)

polygon(c(days, rev(days)),
        c(restaurant + supermarket,
          rev(restaurant + supermarket + household)),
        col = area_colors[3],
        border = NA)

legend("topleft",
       legend = c("Restaurant", "Supermarket", "Household"),
       fill = area_colors,
       bty = "n")

box()
```


mm - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled6* Untitled7* Untitled8* Untitled9* Untitled11* Untitled10* Untitled12*
Source on Save
Run
Source
4 days <- 1:7
5
6 restaurant <- c(25, 30, 28, 35, 32, 38, 34)
7 supermarket <- c(18, 22, 20, 24, 26, 23, 28)
8 household <- c(12, 15, 14, 18, 16, 20, 19)
9
10 waste_data <- rbind(restaurant, supermarket, household)
11
12 # Create stacked area chart
13 area_colors <- c("tomato", "deepskyblue", "gold")
14
15 matplot(days, t(waste_data),
16 type = "n",
17 ylim = c(0, 100),
18 xaxt = "n",
19 main = "Food Waste Across Different Settings",
20 xlab = "Day",
21 ylab = "Food Waste (kg)")
22
23 axis(1, at = days, labels = paste("Day", days))
24
(Top Level)
R Script
Console Terminal Background Jobs
R 4.4.1 · ~/mm/
+ c(restaurant, rev(restaurant + supermarket)),
+ col = area_colors[2],
+ border = NA)
> polygon(c(days, rev(days)),
+ c(restaurant + supermarket,
+ rev(restaurant + supermarket + household)),
+ col = area_color .... [TRUNCATED]
> legend("topleft",
+ legend = c("Restaurant", "Supermarket", "Household"),
+ fill = area_colors,
+ bty = "n")
> box()
Environment History Connections Tutorial
Import Dataset
121 MiB
List
R
Global Environment
Files Plots Packages Help Viewer Presentation
Zoom
Export
Publish
Food Waste Across Different Settings
Restaurant
Supermarket
Household
Food Waste (kg)
0
20
40
60
80
100
Day 1
Day 2
Day 3
Day 4
Day 5
Day 6
Day 7
Day